Subject: PROBABILITY & STATISTICS

Subject Code: 23AS4401

UNIT II : Probability

Syllabus: Probability, probability axioms, addition law and multiplicative law, conditional probability, Bayes' theorem, random variables (discrete and continuous), probability density functions, properties, mathematical expectation.

I. Basic Terms of Probability

Definition:

Random experiment: an experiment whose outcome cannot be predicted in advance, though all possible outcomes are known (tossing a coin, throwing a die).

- ***Sample space (S)***: set of all possible outcomes. Coin: S = {H, T}. Die: S = {1, 2, 3, 4, 5, 6}.
- ***Sample point***: each element of S. ***Event***: any subset of S.
- Simple (elementary) event: one sample point; compound event: more than one.
- ***Sure event*** = S; ***impossible event*** = φ.

Types of events

- ***Mutually exclusive***: cannot occur together, $A \cap B = \varphi$.
- ***Exhaustive***: together cover S, $A \cup B \cup \ldots = S$.
- ***Equally likely***: no outcome is preferred over another.
- ***Independent***: occurrence of one does not affect the other.
- ***Complementary***: A' (or $\bar{A}$) = S – A, "not A".

Set operations as events

Notation	Meaning

$A \cup B$	A or B (at least one) occurs
$A \cap B$	both A and B occur
A'	A does not occur
$A \cap B'$	A occurs but B does not
$A' \cap B'$	neither A nor B occurs

Note:
De Morgan's laws: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$.

2. Definitions of Probability

(a) Classical (Laplace) definition

Formula:
$P(A)$ = (number of favourable cases) / (total number of equally likely cases) = m / n

- Limitations: needs finite and equally likely outcomes.

(b) Statistical (empirical) definition

Formula:
$P(A) = \lim (n \to \infty)$ of m / n (relative frequency in repeated trials)

(c) Axiomatic definition (Kolmogorov, 1933)

Definition:

Let S be a sample space and P a function defined on events. P is a probability if it satisfies the three axioms:

Axiom 1: $P(A) \geq 0$ for every event A (non-negativity)

Axiom 2: $P(S) = 1$ (certainty)

Axiom 3: for mutually exclusive A1, A2, ... : $P(A1 \cup A2 \cup \ldots) = P(A1) + P(A2) + \ldots$

Counting rules (useful for equally likely cases)

Formula:

Permutations: $nPr = n! / (n - r)!$ Combinations: $nCr = n! / [r!(n - r)!]$

- Multiplication rule: if one job can be done in m ways and another in n ways, both can be done in $m \times n$ ways.

Example:

Two dice are thrown. Find P(sum = 7).

Total outcomes = 36. Favourable: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) = 6.

$P = 6/36 = 1/6$

Example:

A committee of 3 is chosen from 5 men and 4 women. P(exactly 2 women)?

$P = (4C2 \times 5C1) / 9C3 = (6 \times 5) / 84 = 30/84 = 5/14$

3. Theorems from the Axioms

- **Theorem 1:** $P(\varphi) = 0$.
  - Proof: $S = S \cup \varphi \cup \varphi \ldots$, all disjoint, so $1 = 1 + P(\varphi) + P(\varphi) + \ldots \Rightarrow P(\varphi) = 0$.

- **Theorem 2 (complement):** $P(A') = 1 - P(A)$.
  - Proof: A and A' are disjoint with $A \cup A' = S \Rightarrow P(A) + P(A') = 1$.
- **Theorem 3:** $0 \leq P(A) \leq 1$.
- **Theorem 4 (monotone):** if $A \subset B$ then $P(A) \leq P(B)$ and $P(B - A) = P(B) - P(A)$.
- **Theorem 5:** $P(A \cap B') = P(A) - P(A \cap B)$.

> **Note:**
> "At least one" problems: P(at least one) = 1 - P(none). It saves a lot of work.

4. Addition Law of Probability

> **Formula:**
> $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- Proof idea: $A \cup B = A \cup (B \cap A')$, disjoint. $P(B) = P(A \cap B) + P(B \cap A')$. Substitute.
- If A, B are mutually exclusive: $P(A \cup B) = P(A) + P(B)$.

For three events

> **Formula:**
> $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

- Boole's inequality: $P(A \cup B) \leq P(A) + P(B)$.

Example:

A card is drawn from a pack of 52. Find P(king or heart).

P(K) = 4/52, P(H) = 13/52, P(K ∩ H) = 1/52

P = 4/52 + 13/52 − 1/52 = 16/52 = **4/13**

Example:

If P(A) = 0.5, P(B) = 0.4, P(A ∩ B) = 0.2 find P(A ∪ B) and P(neither).

P(A ∪ B) = 0.5 + 0.4 − 0.2 = 0.7 ; P(neither) = 1 − 0.7 = **0.3**

5. Conditional Probability and Multiplication Law

Definition:

The conditional probability of A given that B has occurred is

P(A | B) = P(A ∩ B) / P(B), provided P(B) > 0.

- Similarly P(B | A) = P(A ∩ B) / P(A).
- It reduces the sample space to B.

Multiplication law

Formula:

P(A ∩ B) = P(A) · P(B | A) = P(B) · P(A | B)

P(A ∩ B ∩ C) = P(A) · P(B | A) · P(C | A ∩ B)

Independent events

Definition:

A and B are independent if P(A ∩ B) = P(A) · P(B). (equivalently P(A | B) = P(A))

- If A, B independent, then so are (A, B'), (A', B), (A', B').

- Pairwise independence does not imply mutual independence: for 3 events, need $P(A \cap B \cap C) = P(A)P(B)P(C)$ as well.

Note:

Mutually exclusive $\neq$ independent. If $P(A), P(B) > 0$ and $A \cap B = \varphi$ then they are dependent.

Example:

Urn has 5 red and 3 black balls. Two balls are drawn without replacement. P(both red)?

$P = 5/8 \times 4/7 = 20/56 =$ **5/14** (with replacement it would be $5/8 \times 5/8 = 25/64$)

Example:

Three persons shoot at a target independently with probabilities 1/2, 1/3, 1/4. P(target hit)?

P(none hits) $= 1/2 \times 2/3 \times 3/4 = 1/4 \Rightarrow$ P(at least one hits) $= 1 - 1/4 =$ **3/4**

Example:

If $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Are A and B independent?

$P(A)P(B) = 0.2 = P(A \cap B) \Rightarrow$ ***independent***. $P(A \mid B) = 0.2/0.4 = 0.5 = P(A)$.

6. Total Probability and Bayes' Theorem

Definition:

Partition: events B1, B2, ..., Bn are mutually exclusive and exhaustive (B1 ∪ ... ∪ Bn = S) with P(Bi) > 0.

Theorem of total probability

Formula:

$P(A) = \sum P(B_i) \cdot P(A \mid B_i)$ (i = 1 to n)

- Proof: $A = (A \cap B1) \cup ... \cup (A \cap Bn)$, disjoint; apply multiplication law to each term.

Bayes' theorem

Formula:

$P(B_k \mid A) = P(B_k) \cdot P(A \mid B_k) / \sum P(B_i) \cdot P(A \mid B_i)$

- P(Bi) are ***prior*** probabilities; P(Bk | A) is the ***posterior*** probability (revised after A is observed).
- Proof: $P(B_k \mid A) = P(A \cap B_k)/P(A)$; numerator = $P(B_k)P(A \mid B_k)$; denominator by total probability.

Steps to solve Bayes problems

- 1) Name the causes B1, B2, ... and the effect A. 2) Write priors P(Bi).
- 3) Write likelihoods P(A | Bi). 4) Compute P(A) by total probability. 5) Divide.

Example:

Three factories A, B, C make 50 %, 30 %, 20 % of items with defect rates 1 %, 2 %, 3 %. An item is found defective. Find the probability it came from each factory.

Factory	P(Bi)	P(D given Bi)	Product
A	0.5	0.01	0.005

B	0.3	0.02	0.006
C	0.2	0.03	0.006

- $P(D) = 0.005 + 0.006 + 0.006 = \mathbf{0.017}$
- $P(A \mid D) = 0.005/0.017 = \mathbf{5/17} \approx \mathbf{0.294}$; $P(B \mid D) = 6/17 \approx 0.353$; $P(C \mid D) = 6/17 \approx 0.353$ (sum = 1 ✓)

Example:

A disease affects 1 % of people. A test is positive for 95 % of sick and 5 % of healthy people. A person tests positive. P(sick)?

$P(+) = 0.01 \times 0.95 + 0.99 \times 0.05 = 0.0095 + 0.0495 = 0.059$

$P(\text{sick} \mid +) = 0.0095 / 0.059 \approx \mathbf{0.161}$ (only 16 %, because the disease is rare)

7. Random Variables

Definition:

A random variable (r.v.) X is a real-valued function defined on the sample space: $X : S \to R$. It assigns a number to every outcome.

- Example: toss 2 coins, X = number of heads; X takes values 0, 1, 2.
- Two types: ***discrete*** (countable values) and ***continuous*** (any value in an interval).

Discrete random variable

- Takes finite or countably infinite values x1, x2, … (number of defects, number of calls).

Definition:

Probability mass function (p.m.f.): $p(x_i) = P(X = x_i)$

Formula:

Properties: (i) $p(x_i) \geq 0$ (ii) $\Sigma p(x_i) = 1$

Formula:

Distribution function (c.d.f.): $F(x) = P(X \leq x) = \Sigma p(x_i)$ over $x_i \leq x$

- F is non-decreasing, step function, right continuous, $F(-\infty) = 0$, $F(\infty) = 1$.
- $P(a < X \leq b) = F(b) - F(a)$; $p(x_i) = F(x_i) - F(x_{i-1})$.

Example:

Two coins tossed, X = number of heads.

X	0	1	2
p(x)	1/4	1/2	1/4
F(x)	1/4	3/4	1

Example:

Let $p(x) = kx$ for $x = 1, 2, 3, 4$. Find k, $P(X \geq 3)$.

$\Sigma p = k(1 + 2 + 3 + 4) = 10k = 1 \Rightarrow k = 0.1$

$P(X \geq 3) = 0.3 + 0.4 = \mathbf{0.7}$

8. Continuous Random Variable and p.d.f.

Definition:

X is continuous if it takes values in an interval and there is a function f(x), the probability density function (p.d.f.), such that

$P(a \leq X \leq b) = \int f(x)\,dx$ from a to b.

Formula:

Properties of p.d.f.: (i) $f(x) \geq 0$ for all x (ii) $\int f(x)\,dx = 1$ over $(-\infty, \infty)$

- For a continuous r.v., $P(X = c) = 0$, hence $P(a \leq X \leq b) = P(a < X < b) = P(a \leq X < b)$ ...
- $f(x)$ is not a probability; it is a density (can exceed 1).

Formula:

c.d.f.: $F(x) = P(X \leq x) = \int f(t)\,dt$ from $-\infty$ to x; $f(x) = dF(x)/dx$

- F is continuous, non-decreasing, $0 \leq F \leq 1$, $F(-\infty) = 0$, $F(+\infty) = 1$.
- $P(a < X < b) = F(b) - F(a)$.
- Median M: $F(M) = 1/2$. Mode: x where $f(x)$ is maximum. Quartiles: $F(Q1) = 1/4$, $F(Q3) = 3/4$.

Example:

Let $f(x) = kx^2$ for $0 \leq x \leq 3$, 0 otherwise. Find k, $P(1 < X < 2)$, $F(x)$.

$\int kx^2\,dx$ from 0 to 3 $= k \times 27/3 = 9k = 1 \Rightarrow$ **$k = 1/9$**

$P(1 < X < 2) = (1/9) \times (8 - 1)/3 =$ **$7/27$**

$F(x) = x^3/27$ for $0 \leq x \leq 3$; 0 for $x < 0$; 1 for $x > 3$.

Example:

A r.v. has $f(x) = 2x$, $0 < x < 1$. Find the median.

$F(M) = M^2 = 1/2 \Rightarrow M = 1/\sqrt{2} \approx$ **0.707**

Example:

Given $f(x) = 3(1 - x)^2$ on $(0, 1)$. Check it is a valid p.d.f.

$\int 3(1 - x)^2\,dx = [-(1 - x)^3]$ from 0 to 1 $= 0 + 1 = 1$ ✓, and $f \geq 0 \Rightarrow$ valid.

9. Mathematical Expectation

Definition:

Expectation (mean) of a r.v. is its long-run average value.

Formula:

Discrete: $E(X) = \Sigma\, x_i \cdot p(x_i)$

Formula:

Continuous: $E(X) = \int x\, f(x)\, dx$ over $(-\infty, \infty)$

Formula:

Function of X: $E[g(X)] = \Sigma\, g(x_i)\, p(x_i)$ or $\int g(x)\, f(x)\, dx$

Properties of expectation

- $E(c) = c$; $E(aX + b) = a\,E(X) + b$.
- $E(X + Y) = E(X) + E(Y)$ (always).
- $E(XY) = E(X)\,E(Y)$ if X, Y are independent.
- If $X \geq 0$ then $E(X) \geq 0$; if $X \leq Y$ then $E(X) \leq E(Y)$.

Variance

Formula:

$Var(X) = E[(X - \mu)^2] = E(X^2) - [E(X)]^2$, $SD = \sqrt{Var}$

- $Var(c) = 0$; $Var(aX + b) = a^2\, Var(X)$.
- $Var(X \pm Y) = Var(X) + Var(Y)$ if X, Y independent.

Moments and M.G.F.

- r-th raw moment $\mu_r' = E(X^r)$; central moment $\mu_r = E[(X - \mu)^r]$.

Formula:

M.G.F.: $M_X(t) = E(e^{tX})$; $E(X^r) = d^rM/dt^r$ at $t = 0$

- $M_{aX+b}(t) = e^{bt} M_X(at)$; for independent X, Y: $M_{X+Y} = M_X \cdot M_Y$.

Example:

Find E(X) and Var(X) for $p(x) = 0.1x$, $x = 1, 2, 3, 4$.

$E(X) = 0.1(1 + 4 + 9 + 16) = \mathbf{3}$; $E(X^2) = 0.1(1 + 8 + 27 + 64) = 10$

$Var(X) = 10 - 3^2 = \mathbf{1}$

Example:

For $f(x) = x^2/9$ on (0, 3) find mean and variance.

$E(X) = \int x^3/9\, dx = 81/36 = \mathbf{9/4}$; $E(X^2) = \int x^4/9\, dx = 243/45 = 27/5$

$Var(X) = 27/5 - 81/16 = (432 - 405)/80 = \mathbf{27/80} = 0.3375$

Example:

A game: win Rs 100 with probability 0.2, lose Rs 20 with probability 0.8. Expected gain?

$E = 100(0.2) + (-20)(0.8) = 20 - 16 = \mathbf{Rs\ 4}$ per play (favourable game).

Example:

For $f(x) = 2x$ on (0,1): $E(X) = \int 2x^2\, dx = 2/3$, $E(X^2) = \int 2x^3\, dx = 1/2$, Var $= 1/2 - 4/9 = \mathbf{1/18}$.

10. More Worked Examples

Probability basics

Example:

Three coins are tossed. Find P(at least two heads).

S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}, n(S) = 8.

Favourable: HHH, HHT, HTH, THH = 4 ⇒ P = 4/8 = **1/2**

Example:

Two cards are drawn from a pack without replacement. P(both aces)?

P = 4C2 / 52C2 = 6/1326 = **1/221** (or 4/52 × 3/51)

Example:

P(A) = 1/4, P(B) = 1/3 and P(A ∪ B) = 1/2. Are A, B independent?

P(A ∩ B) = 1/4 + 1/3 – 1/2 = 1/12 = P(A)P(B) ⇒ ***independent***; P(A | B) = (1/12)/(1/3) = 1/4.

Example:

A box has 4 red and 6 blue balls. Two are drawn. P(same colour)?

P = (4C2 + 6C2) / 10C2 = (6 + 15)/45 = **7/15**

Example:

Two students pass independently with probabilities 0.7 and 0.6. Find P(exactly one passes) and P(at least one passes).

P(exactly one) = 0.7 × 0.4 + 0.3 × 0.6 = 0.28 + 0.18 = **0.46**

P(at least one) = 1 – 0.3 × 0.4 = **0.88**

Bayes' theorem

Example:

Bag I has 3 white, 2 black balls; Bag II has 2 white, 4 black. A bag is chosen at random and a ball drawn is white. P(it came from Bag I)?

P(W) = 1/2 × 3/5 + 1/2 × 2/6 = 3/10 + 1/6 = 14/30 = 7/15

P(I | W) = (3/10) / (7/15) = **9/14**

Example:

60 % of students are boys, 40 % girls. 10 % of boys and 5 % of girls are left handed. A left-handed student is chosen. P(girl)?

P(girl | L) = (0.4 × 0.05) / (0.6 × 0.10 + 0.4 × 0.05) = 0.02 / 0.08 = **0.25**

Random variables

Example:

Three coins tossed, X = number of heads. Find the p.m.f., E(X), Var(X).

X	0	1	2	3
p(x)	1/8	3/8	3/8	1/8

- $E(X) = (0 + 3 + 6 + 3)/8 = \mathbf{1.5}$; $E(X^2) = (0 + 3 + 12 + 9)/8 = 3$
- $Var(X) = 3 - 2.25 = \mathbf{0.75}$

Example:

A fair die is thrown. Find E(X) and Var(X).

$E(X) = (1+2+3+4+5+6)/6 = \mathbf{3.5}$; $E(X^2) = 91/6$

$Var(X) = 91/6 - 49/4 = (182 - 147)/12 = \mathbf{35/12} \approx \mathbf{2.92}$

Example:

X takes -1, 0, 1, 2 with probabilities 0.2, 0.3, 0.4, 0.1. Find E(X), Var(X), E(2X + 3), Var(2X + 3).

$E(X) = -0.2 + 0 + 0.4 + 0.2 = \mathbf{0.4}$; $E(X^2) = 0.2 + 0 + 0.4 + 0.4 = 1.0$

$Var(X) = 1 - 0.16 = \mathbf{0.84}$

$E(2X + 3) = 2(0.4) + 3 = \mathbf{3.8}$; $Var(2X + 3) = 4 \times 0.84 = \mathbf{3.36}$

Example:

Sum of two dice: E(sum) = E(X1) + E(X2) = 3.5 + 3.5 = **7** (by linearity of expectation).

Var(sum) = 35/12 + 35/12 = 35/6 (independent dice).

Continuous random variables

Example:

$f(x) = e^{-x}$, $x > 0$. Find $P(X > 2)$, mean, variance, median.

$P(X > 2) = \int e^{-x}\,dx$ from 2 to $\infty = e^{-2} \approx$ **0.1353**

$E(X) = \int x\,e^{-x}\,dx = 1$; $E(X^2) = \int x^2 e^{-x}\,dx = 2 \Rightarrow$ Var = 2 – 1 = **1**

Median: $1 - e^{-M} = 1/2 \Rightarrow M = \ln 2 \approx$ **0.693**

Example:

$f(x) = k(x + 1)$ on (0, 2). Find k, $P(X < 1)$, $E(X)$.

$\int k(x + 1)\,dx = k(2 + 2) = 4k = 1 \Rightarrow$ **k = 1/4**

$P(X < 1) = (1/4)(1/2 + 1) =$ **3/8**

$E(X) = (1/4) \int (x^2 + x)\,dx = (1/4)(8/3 + 2) =$ **7/6**

Example:

M.G.F. of a Bernoulli variable ($P(X=1) = p$, $P(X=0) = q$):

$M(t) = q + p\,e^t \Rightarrow M'(0) = p = E(X)$; $M''(0) = p = E(X^2) \Rightarrow$ Var = $p - p^2 =$ **pq**

II. Practice Problems with Answers

- 1) $P(A) = 0.6$, $P(B) = 0.5$, $P(A \cup B) = 0.8$. Find $P(A \cap B)$ and $P(A \mid B)$.
  - Ans: $P(A \cap B) = 0.3$; $P(A \mid B) = 0.3/0.5 = 0.6$.
- 2) $f(x) = kx(1 - x)$, $0 < x < 1$. Find k, mean, variance.

- Ans: k = 6; mean = 1/2; $E(X^2) = 6(1/4 - 1/5) = 0.3$; variance = 0.3 − 0.25 = 0.05.

- 3) $p(x) = c(2x + 1)$, $x = 0, 1, 2, 3, 4$. Find c, $P(X < 2)$, $E(X)$.
  - Ans: $25c = 1 \Rightarrow c = 1/25$; $P(X < 2) = (1 + 3)/25 = 4/25$; $E(X) = (0+3+10+21+36)/25 = 2.8$.
- 4) A, B independent with $P(A) = 0.3$, $P(B) = 0.4$. Find $P(A \cup B)$, $P(A' \cap B')$.
  - Ans: $0.3 + 0.4 - 0.12 = 0.58$; $P(A' \cap B') = 0.7 \times 0.6 = 0.42$.
- 5) Prove $P(A \cap B') = P(A) - P(A \cap B)$.
  - Hint: $A = (A \cap B) \cup (A \cap B')$, disjoint; apply axiom 3.
- 6) Show $Var(aX + b) = a^2 Var(X)$.
  - Hint: $E(aX + b) = a\mu + b$, so $(aX + b - a\mu - b)^2 = a^2(X - \mu)^2$.

12. Formula Sheet – Unit II

Topic	Formula
Classical	$P(A) = m/n$
Complement	$P(A') = 1 - P(A)$
Addition	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Conditional	$P(A \mid B) = P(A \cap B)/P(B)$
Multiplication	$P(A \cap B) = P(A)P(B \mid A)$
Total prob.	$P(A) = \Sigma P(Bi)P(A \mid Bi)$
Bayes	$P(Bk \mid A) = P(Bk)P(A \mid Bk)/P(A)$
Mean	$E(X) = \Sigma xp$ or $\int xf\,dx$
Variance	$E(X^2) - [E(X)]^2$

Note:
Always draw a tree diagram for Bayes problems – multiply along branches, add across branches.

Quick Revision - Unit II

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $P(A \mid B) = P(A \cap B)/P(B)$.
- Independent $\Leftrightarrow P(A \cap B) = P(A)P(B)$.
- Bayes: $P(Bk \mid A) = P(Bk)P(A \mid Bk) / \Sigma P(Bi)P(A \mid Bi)$.
- Discrete: $\Sigma p = 1$; continuous: $\int f = 1$; $F' = f$.
- $E(X) = \Sigma xp$ or $\int xf$; $Var = E(X^2) - [E(X)]^2$.